Online Resource 1 for: Deep Reinforcement Learning for the Interval Job Shop Scheduling Problem: A Comparison with Genetic Programming Hyper-Heuristics across Inference Budgets

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Abstract

This resource contains material referenced from the paper: the automatic hyperparameter-configuration campaigns and their search spaces (Section 1), the architecture and controlled ablation of the self-attention variant (Section 2), and the per-instance results on the 70-instance benchmark (Section 3). Section, table and equation numbers of the form S$\boldsymbol{n}$ refer to this document; all other references are to the paper.

1 The hyperparameter-configuration campaigns

The values of Table 5 of the paper were fixed by hand during development, most of them at the defaults customary in the PPO literature. To assess whether automatic tuning would improve on them, two configuration studies were run with irace (López-Ibáñez, Dubois-Lacoste, Pérez Cáceres, Birattari, & Stützle, 2016). The first raced 330 experiments (309 used) over the 8-parameter space of Table 1 at reduced fidelity: one training instance, 300 episodes, ∼8 minutes per evaluation. Its elites agreed on a more aggressive update regime and kept minibatch size and update period at their defaults, but at the full 1000-episode operating budget the best of them scored 14.5% mean RE against 13.4% for the defaults: the low-fidelity optimum did not transfer.

Table 1 Search space of the two irace campaigns. Real parameters were sampled on a logarithmic scale, the rest from the listed values.

Parameter	Default	Campaign 1 (330 exp.)	Campaign 2 (300 exp.)
learning rate	$3 \cdot 10^{-4}$	$[10^{-4}, 10^{-3}]$	$[10^{-4}, 10^{-3}]$
entropy coefficient	0.01	[0.003, 0.03]	[0.003, 0.03]
clipping range	0.2	0.1, 0.2, 0.3	0.1, 0.2, 0.3
epochs per update	4	2, 4, 8	4, 8
GAE λ	0.95	0.90, 0.95, 0.98	0.90, 0.95, 0.98
hidden width	128	64, 128, 256	128, 256
minibatch size	256	128, 256, 512	fixed
update period	4 ep.	2, 4, 8 ep.	fixed

The second campaign raced 295 experiments at operating fidelity, with 1000 episodes on two instances per evaluation, over the reduced 6-parameter space of Table 1, seeded with the defaults and the first campaign's elites. It converged to a region distinct from the defaults and eliminated the seeded default during racing; yet its winner, retrained on the four training instances over three seeds, again failed to improve: 14.73% against 13.52% over the six validation instances. A third campaign raced the reward instead of the optimizer: the six weights of Eq. (5) of the paper, each free in $[0, 1]$, over 299 experiments at operating fidelity. Its winner did not improve the deployed weights either, 14.04% against 13.55% under the same confirmation protocol (Wilcoxon signed-rank, $p = 0.21$). Two by-products of that campaign are more informative than its verdict. A seeded terminal-only vector $(1, 0, 0, 0, 0, 0)$ was eliminated in the first round, so the dense shaping is not dispensable; and four elites survived the full race differing by as much as 0.48 in a single weight, with a Kendall agreement across seeds of $W = 0.02$–0.05 on which is better, so at this budget the objective is flat in the reward weights relative to the noise between training runs.

Automatic configuration therefore does not improve on the hand-set values for this problem, even at the operating budget, nor do raced reward weights improve on the hand-set ones, within the resolution of the protocol. That resolution can be quantified: with a standard deviation of 0.47 points across seed means at 1000 episodes and three seeds per arm, the minimum difference a paired confirmation detects at conventional power is approximately one point of RE. The three campaigns therefore exclude improvements larger than that magnitude, not the existence of smaller ones. The same arithmetic explains their common signature, the racing optimum never matching the confirmation optimum: single-run outcomes are too noisy to separate near-equivalent configurations, and raising the resolution by repeating each evaluation reduces the number of configurations a fixed budget can race.

2 The self-attention variant

Pooling summarizes the candidate set by two statistics, against which every candidate is then scored. This construction cannot express a comparison between two *particular* candidates: whether operation i should take the machine now may depend on which other eligible operation would occupy it next, and the mean embedding does not identify that candidate. The embeddings of all $|\mathcal{E}|$ candidates are available at the decision step, so what the architecture lacks is not the information but a path along which one candidate can condition another. *Self-attention* (Vaswani et al., 2017) supplies such a path and is the component most commonly adopted for this purpose in the neural combinatorial optimization literature. In an attention layer every candidate emits a *query*, a *key* and a *value*; the similarity between one candidate's query and another's key decides how much of that other's value is added into the first's embedding. Each candidate is thereby rewritten as a content-weighted reading of all the others, with the weights computed from the embeddings themselves rather than fixed in advance.

A stack of B pre-layer-normalization (pre-LN) transformer blocks transforms the embeddings between the embedding and pooling stages of the base architecture (Figure 1 of the paper). Writing $\Phi \in \mathbb{R}^{|\mathcal{E}| \times h}$ for the stacked embeddings, each block computes

$$Z = \Phi + \mathrm{MHA}\big(\mathrm{LN}(\Phi)\big), \qquad \Phi' = Z + \mathrm{FFN}\big(\mathrm{LN}(Z)\big), \tag{1}$$

where MHA is multi-head scaled dot-product attention (4 heads of dimension 32, no dropout), FFN is a two-layer perceptron ($128 \to 256 \to 128$, ReLU) applied to each row, and LN is LayerNorm. Both additions are *residual*, that is, each block adds to its input rather than replacing it, so an untrained block is close to the identity and the stack starts near the base model; the pre-LN arrangement is preferred over the original post-LN one for this property (Xiong et al., 2020). No positional encoding is used, which keeps self-attention permutation-equivariant, so the pooled context remains permutation-invariant and the size invariance of the base architecture is preserved. Padding slots are masked out of the attention weights, and unit tests verify that masked-padded forward passes are numerically identical to unpadded ones.

The variant is a configuration option on the same code path, and $B{=}0$ reproduces the base network exactly, so the comparison below differs in this component and in nothing else. A single block carries 132,480 parameters in the query, key and value projections, the output projection, two layer normalizations and the feed-forward network, so the $B{=}2$ variant evaluated here grows the network from 1.2×10^5 to 3.9×10^5 parameters, a factor of 3.2, and multiplies the wall-clock cost of an episode by 2.2.

The controlled ablation.

The variant inserts two residual attention blocks between the shared per-operation encoder and the masked pooling, which it leaves in place; everything else is held fixed, including the training and evaluation protocols, the same four training instances, the same six validation instances, the same seeds and the hyperparameters of Table 5 of the paper. The two arms are therefore matched instance by instance and seed by seed, and analysed with the instance as the experimental unit, at both training budgets employed in the study: over three seeds at 300 episodes and over ten at 1000, the operating budget.

The variant yields no improvement at either budget and a significant degradation at the operating one. At 300 episodes it is 1.03 points of RE inferior (95% CI 0.01 to 2.06), degraded on 5 of the 6 instances, a difference that does not reach significance (Wilcoxon signed-rank, $p = 0.063$); at 1000 episodes the deficit is 1.06 points (0.46

to 1.65), degraded on all six instances and in every one of the ten seeds ($p = 0.031$, the smallest value six instances admit). Table 2 reports the corresponding means and best seeds.

Table 2 Attention ablation: mean [best seed] RE (%) on the validation set, over three seeds at 300 episodes and ten at 1000.

Configuration	300 eps	1000 eps
Deep Sets	**16.6** [16.2]	**13.8** [13.1]
+ attention (B=2)	17.6 [17.2]	14.9 [14.2]

The representation accounts for part of this outcome. One of the per-operation features is itself relational with respect to the eligible set: slack measures the candidate's earliest start against the earliest among all candidates, so the comparison an attention layer would have to learn along that axis is already available as a precomputed input. A second, machine congestion, is relational with respect to the machines rather than the candidates, contrasting a machine's remaining load with the average over machines. What remains are the pairwise relations these aggregates do not summarize, and at the budgets tested they do not amortize their cost; larger training budgets or more heterogeneous instance distributions may be required for such structure to become profitable.

3 Per-instance results

Table 3 lists, for each of the 70 interval Taillard instances, the crisp lower bound and the RE (%) of the earliest-start rule, of Giffler–Thompson with MWKR tie-breaking, of the evolved rule of Díaz Rodríguez (2026), and of the policy at one pass and at 1024 samples, the artifact each study selects as in Table 8 of the paper. The ten 20×15 instances TA11–TA20 are the training and validation set.

References

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Table 3 Per-instance RE (%) of the constructive methods and of the policy. LB is the crisp Taillard lower bound; GP the evolved rule; Pol. the policy at one pass and Pol.* at 1024 samples.

Inst.	LB	EST	G&T	GP	Pol.	Pol.*	Inst.	LB	EST	G&T	GP	Pol.	Pol.*
TA1	1231	49.9	26.1	19.3	17.2	12.3	TA36	1819	41.1	42.8	23.8	19.2	13.4
TA2	1244	24.9	27.8	14.2	14.8	7.9	TA37	1771	43.4	35.6	18.4	22.4	17.7
TA3	1218	33.0	33.2	14.1	14.0	8.2	TA38	1673	45.0	28.0	20.1	20.1	15.7
TA4	1175	32.0	27.7	13.0	18.5	11.3	TA39	1795	46.6	28.9	22.4	22.3	14.7
TA5	1224	27.2	33.0	14.7	10.6	7.8	TA40	1651	46.9	42.3	26.1	27.8	19.4
TA6	1238	26.8	18.2	13.7	14.8	7.7	TA41	1906	44.6	45.0	33.4	27.8	24.3
TA7	1227	30.7	26.1	17.7	15.2	9.1	TA42	1884	71.1	38.1	24.3	23.7	20.3
TA8	1217	23.7	25.0	14.8	17.3	9.8	TA43	1809	68.7	39.8	22.3	27.4	20.7
TA9	1274	37.7	28.2	20.0	12.9	11.1	TA44	1948	52.1	34.7	25.8	27.4	17.9
TA10	1241	37.0	22.1	15.4	15.1	7.6	TA45	1997	50.2	33.7	14.9	16.7	13.0
TA11	1357	56.2	30.7	17.5	19.8	12.4	TA46	1957	60.8	31.6	22.4	21.7	19.6
TA12	1367	48.8	34.6	17.6	13.0	11.1	TA47	1807	46.8	30.1	25.6	24.4	19.8
TA13	1342	60.2	26.9	13.3	19.7	12.0	TA48	1912	59.1	33.7	21.1	22.8	19.3
TA14	1345	39.5	29.7	13.2	17.7	9.0	TA49	1931	51.0	35.3	20.1	23.3	18.7
TA15	1339	48.1	30.6	16.7	17.0	11.6	TA50	1833	61.4	46.2	31.3	32.2	25.6
TA16	1360	45.2	34.0	15.6	15.9	9.3	TA51	2760	33.8	34.7	23.2	24.3	13.4
TA17	1462	58.9	17.7	13.9	14.5	9.9	TA52	2756	38.0	26.5	12.7	19.0	12.7
TA18	1377	43.8	31.8	23.6	20.9	14.1	TA53	2717	34.1	18.3	11.8	13.5	9.7
TA19	1332	45.8	29.3	20.0	14.8	10.0	TA54	2839	20.9	16.1	8.2	9.9	5.2
TA20	1348	45.7	24.1	15.1	20.7	12.8	TA55	2679	31.7	29.2	15.5	20.0	14.4
TA21	1642	33.4	28.0	20.2	17.6	11.2	TA56	2781	34.8	19.8	10.9	10.9	9.2
TA22	1561	44.2	30.8	17.4	17.7	15.4	TA57	2943	32.9	20.0	9.6	15.3	10.1
TA23	1518	42.8	35.3	17.8	20.2	12.2	TA58	2885	37.4	26.3	13.6	15.3	10.1
TA24	1644	56.3	23.8	14.1	13.4	11.0	TA59	2655	26.6	28.2	16.7	21.5	15.0
TA25	1558	45.8	33.8	16.6	18.3	11.0	TA60	2723	41.7	19.8	15.5	11.1	9.0
TA26	1591	42.8	31.2	26.9	18.8	13.1	TA61	2868	45.7	25.4	15.4	18.2	15.2
TA27	1652	50.0	31.1	21.4	18.3	13.2	TA62	2869	43.0	27.6	17.2	20.2	15.9
TA28	1603	36.8	26.8	12.4	15.6	9.3	TA63	2755	38.8	23.8	13.4	17.8	12.3
TA29	1583	28.8	32.0	17.2	14.9	9.8	TA64	2702	38.2	28.4	12.2	13.2	12.2
TA30	1528	34.2	36.1	17.0	21.9	15.6	TA65	2725	42.9	30.0	20.5	20.8	17.3
TA31	1764	30.7	34.4	24.5	20.5	15.9	TA66	2845	41.8	28.0	11.9	15.7	13.2
TA32	1774	42.7	33.3	19.0	25.1	18.1	TA67	2825	31.7	26.1	15.2	15.9	13.0
TA33	1788	51.3	34.8	27.5	30.7	21.0	TA68	2784	50.1	19.1	10.6	15.4	10.6
TA34	1828	45.9	31.8	20.7	18.2	16.0	TA69	3071	35.3	22.8	12.9	15.4	11.6
TA35	2007	30.3	24.1	8.4	7.9	4.1	TA70	2995	42.9	23.4	16.7	18.4	15.3

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